Fear Effects

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Fear is the player's health. As it rises, the game should *feel* more dangerous. This system turns the player's fear into feedback:

- **Fear Vignette:** a noisy vignette that creeps in from the screen edges.
- **Heartbeat:** louder and faster as fear rises.
- **Intrusive effects:** a ready-made slot for in-world scares (not filled in yet).

The fear value belongs to `FearBar` (part of scoring). This system only reads it.

See also: Shaders & VFX (the Fear Vignette shader), Interface & Settings (Fear Vignette setting), Audio Manager.

How It Fits Together

`FearEffectsManager` reads fear and passes it to the effect components next to it:

Call	When	For
`UpdateIntensity(normalizedFear, isInEncounter)`	Every frame	Effects that scale smoothly (fear from 0 to 1).
`OnRankChanged(rank)`	When `FearBar.OnFearChanged` fires	Effects that change in steps.

Rank	Fear
Healthy	15 or less
Fine	Above 15, up to 40
Critical	Above 40, up to 80

Rank	Fear
Doomed	Above 80

`isInEncounter` comes from `PlayerStatus.IsInEncounter`, a property for whatever starts and ends combat to set. Nothing uses it yet.

`OnFearChanged` fires on **every** fear change, not just rank changes, so rank handlers should be fine being called with the same rank twice.

Setup

1. Put **FearEffectsManager** on an object in the scene.
2. Put the effect components (**FearPostProcessEffects**, **FearAudioEffects**, **FearIntrusiveEffects**) on the **same object**. Any can be left off.
3. Make sure the scene's Volume profile has a **Fear Vignette** override and the Fear Vignette renderer feature is on the renderer.

Fear Bar and Player Status are found automatically if left empty.

Fear Vignette

```
strength = fear (0 to 1) x Effect Strength x player's Fear Vignette setting
```

As strength rises the vignette closes in, its edge gets more broken up by noise, and the noise moves faster.

Field	What it does
Post Process Volume	The Volume with the Fear Vignette override. If empty, uses the first Volume under **SceneEssentialsBundle**.
Vignette Color, **Blend Mode**	The look (Multiply, Screen, Overlay or Hard Light).
Effect Strength	Overall multiplier.
Vignette Intensity Max	How far in it reaches at full fear.
Noise Intensity Max	How broken up the edge gets at full fear.
Noise Speed Min / Max	Creep speed at no fear and full fear.
Vignette Softness, **Noise Scale**, **Cycle Duration**	Passed straight to the shader.

Changing the Fear Vignette graphics setting re-applies the effect straight away, even while paused. Disabling the component turns the vignette off.

> **In code:** values are written with `Override()` on the `FearVignetteVolume` from the Volume's `sharedProfile` (using `profile` would create a copy that never shows on screen).

Heartbeat

Plays at every rank except Healthy. Each rank has its own sound and beat interval, so it speeds up as the rank rises. Volume follows fear smoothly from **Volume Min** to **Volume Max**.

Field	What it does
Fine / Critical / Doomed Beat	SoundDataSO per rank (clip and pitch are used, volume comes from fear).
Fine / Critical / Doomed Interval	Seconds between beats.
Volume Min / Max	Volume at no fear and full fear.

It adds its own AudioSource, routed through the Fine Beat's mixer group.

Adding a Fear Effect

1. Write a component with either or both methods:

```
public void UpdateIntensity(float normalizedFear, bool isInEncounter) { }
public void OnRankChanged(FearBar.FearRank rank) { }
```

2. Put it on the manager's object.
3. In `FearEffectsManager`, add a field, `GetComponent` it in `Awake`, and call it from `Update` and `OnRankChanged`.

`FearIntrusiveEffects.OnRankChanged` and `FearPostProcessEffects.OnRankChanged` are already wired up and empty, ready for in-world scares and rank-based screen effects (e.g. chromatic aberration at Doomed).

If the effect is visual, scale it by a Graphics Settings multiplier so the player can turn it down.

Gotchas

- **Effect components must be on the same object as the manager.**

- **Nothing on screen?** Check the profile has a Fear Vignette override and the renderer has the feature.